

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 1 of 8

(1) Research Program/Project/Study Title: Innovation Ecosystem Assessment in SUCs: A Case of Capiz State University	
Proponent Name:	DR. EFREN L. LINAN
Academic Rank:	Professor III
Co-Authors (with academic rank):	DR. LEO ANDREW B. BICLAR , Associate Professor IV MR. MICHAEL VAUGHN J. CADIZ , CRJ Specialist
Campus/Satellite College/Address:	RDE Office
Telephone/Cell Phone/Fax Number:	
E-mail Address:	ellinan@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
---	--

(4) Site of implementation	
Municipality/City:	ROXAS CITY
Province:	CAPIZ
Region:	VI

(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
--	---

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 2 of 8

(7) Sector/Commodity	(8) Discipline/Field of Specialization Social Science
-----------------------------	---

(9) Rationale

The Philippines is currently in the 51st place in the 2021 Global Innovation Index rankings and is considered a Lower Middle-Income Group country. Based on its accelerated growth of being from the 91st to the 51st rank, the Philippines is considered to be an emerging economy through its efforts in shaping its innovation landscape. Currently, the Philippines is at the top quartile in terms of Knowledge, and Technology, and Business Sophistication but is ranked lower when it comes to its Institutions, Human Capital and Research, Infrastructure, Market Sophistication, and Creative outputs (Dutta et al., 2021). Nevertheless, institutions such as Capiz State University have been addressing and taking action regarding the gaps in order to enhance the different pillars that build up the country's innovation index.

Producing quality research and technology has always been an integral part of the long-term goals of CAPSU. Abundant raw resources such as various seafood produce, crops, native poultry, and livestock are at the disposal of CAPSU's faculty researchers. Thus, with the diversity of readily available resources, it is up to the faculty researchers to innovate around these products and create a production chain that will boost the quality of life, exporting power and livelihood of the Capizeños. Based on the data provided by the Philippine National Statistics Office (n.d.), over 727 farming plots are not being utilized, and 61,219 are used for temporary farming. This could be an opportunity for the researchers of CAPSU to develop viable technologies that could utilize the full potential of unused lands through research-based projects and development.

With that said, in order to be able to provide quality services and results to the community, CAPSU's faculty researchers must be able to produce quality research and possible results that will have a significant impact on the scientific community. Bowsher et al. (2019) highlighted the three main narratives that emerged during their study on developing the health research capacity. These themes were the "effect of power relations on capacity development, stronger links between research, policy and practice, and the importance of systems approach." Although this system of findings may vary from institution to institution, this can be used as a component of determining the research capacity of the institution's faculty researchers.

At the start of every year, the CAPSU Research, Development, and Extension (RDE) Training Office are regularly conducting a Training Needs Assessment (TNA) for its faculty researchers, which assesses the training and capability-building activities that they need in order to enhance their capacity in producing quality research. In hindsight, it is hypothesized that the faculty researchers have a level of awareness of their own capability from conceptualization up to the publication process of their outputs. Nevertheless, research findings regarding the research capability of CAPSU's faculty researchers are still nonexistent. Therefore, the RDE office made this their priority in order to help the faculty researchers achieve their roadmap goal and contribute to the success of their Innovation Ecosystem.

(10) Objectives/Statement of the Problem

This study aims to assess the Innovation Ecosystem of Capiz State University. Specifically, it seeks to achieve the following objectives.

1. Determine the research capabilities and involvement of CAPSU faculty

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 3 of 8

2. Identify the level of the self-assessed research capability of the CAPSU Faculty
3. Assess the status of research utilization.
4. Assess the extent of participation of other stakeholders.
5. Correlation analysis of demographics and other pertinent parameters.
6. Create a roadmap to enhance the Innovation Ecosystem of the University.
7. Formulate an enhanced training program that can address the needs of the faculty researchers in conducting their studies.
8. Identify the possible technology/software that can help the faculty researchers in their research efforts.

(11) Review of Literature/Prior Art Search

Parameters for Creating a Quality Research Paper

Specific parameters must be met in order to create a high-quality research output. As cited by Chu et al. (2014), there are four prerequisites to do so. First is the individual skills of each researcher, which boil down to their ability to write, analyze, synthesize, and utilize their readily available research tools. Next is the appropriate infrastructure, which includes but is not limited to: the work environment, research equipment, laboratory, and others. The third prerequisite is the research's relevance to a nation's short-term and long-term goals and policies. This is one of the main determining factors of whether or not a research proposal will be supported or funded by the researcher's institution or partner agencies. The final prerequisite is the perceived impact or its contribution to the global/international research efforts and needs. It was further stated that the African Researchers could barely reach international standards due to the lack of these prerequisites and support from different funding agencies. Due to these factors, capable researchers are being exploited by other peers, institutions, and agencies, using the "Publish or Perish" mandates of their government as well as the government.

Difficulties of Faculty Researchers in Conducting Research

In the Philippines, pursuing personal and professional growth is a challenge due to heavy workloads, multiple designations, lack of capacity-building activities, and lack of funding (especially for publications) due to the unrevised policies of the government agencies is the most difficult hurdle that a faculty researcher can face (Vecaldo et al., 2019). The "Publish or Perish" mandate also exists in the country as it requires the faculty of Higher Education Institutions (HEIs) to publish in ISI/Scopus-indexed journals and publishers in order to be promoted to higher levels of position classification and to receive compensation (DBM, n.d.).

This puts a lot of unnecessary pressure on the HEI faculty to publish in such journals using outdated methodologies, lack of research proficiency due to poor funding, endangering one's social identity, lack of research proficiency due to heavy workload, and other factors (Kurt, 2018). This is also evident within CAPSU because it is a State University and Colleges (SUCs). Therefore, it is subject to the government's rules, laws, and policies regarding career advancement. It was only until the year 2021 that CAPSU could

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 4 of 8

hire personnel that could handle specific tasks of its faculty researchers that are integral to the success of their paper, such as external and internal publication.

Engaging Faculty Researchers to Conduct Research

According to Huenneke et al. (2017), established research institutions practice adding newly hired greenhorn faculty to existing studies and projects in order for them to learn more in their field and to create research outputs. In their study, the participants had a standard model on how to increase the research activity within their institution. These models are: “cluster hiring, revision of institutional policies and infrastructure, and systematic prioritization of research for administrators as well as for individual faculty members.”

From the local perspective, the revision of policies is currently underway and is the primary objective of the RDE Office. The systematic prioritization had already commenced due to the 5-year road map and innovation ecosystem that was formulated by the entire RDE body.

Based on Huenneke’s findings, investing in institutional equipment, components, and resources or by means of acquiring external funding for institutional development can greatly contribute to its long-term goal than recruiting or hiring new faculty researchers from other institutions. These supported institutions should have a program focused on **career development plans (aligned with research)** for junior and new faculty members. This way, the universities would then have a pool of talented individuals that can greatly contribute to their future research efforts and their country’s national agenda.

In the same study, Huenneke stated that **Transition and Succession planning** should be heavily considered. In relation, CAPSU also changes its assigned leaders annually. This can cause shifts of interest in research agendas, extension programs, and other aspects of the RDE Office. Transition and succession had been a common practice within the institution (through verbal or internal agreement), but no formal policies or advisories exist. This could also be a determining factor to the research capacity of the faculty researchers.

(12) Scientific Basis/Theoretical Framework

This study will be anchored on the study of Tolentino (2021) where he utilized the perceived level of the research capability of their science teachers in basic education using a self-assessment scale adopted from Abrazo & Coronel (2016). The results of their study provided them the insight of what are the existing gaps and concerns they should address in order to enhance the capability of their faculty researchers.

Although, this is only a self-perceived scale, the researchers of this study have also taken into consideration the external perspective in perceiving the research capacity of a researcher in order to have empirical evidence that can be used for additional policy enhancement, capability building activities, program enhancement proposals, and other inputs for the development of the faculty researchers of CAPSU.

Therefore, the analysis of the empirical evidence of this study will be anchored on the study of Kotesmir & Shashnov (2017) to assess the quality of their research output. Within their study, they have analyzed the following aspects: Dynamic Publication Activity, Citation Indicators, the University’s Integration into International Research Collaboration, Individual’s Research Capacity, and the Journal Quality of their published papers. Additionally, components such as research quality, similarity index, and grammar quality will be considered for analysis.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 5 of 8

(13) Methodology

Research Design

This study will be a Mixed Methods Research with the Triangulation Design: Validating Quantitative Data Model Cresswell, 2014). Through this design, the faculty researchers' self-perceived research capacity (Quant) will be further explored through a semi-structured interview regarding their strengths and difficulties in conducting research (Qual). In turn, this could be further validated using their research output, citations, and other research-based activities in order to create pragmatic results of their capacity.

Research Instrument

This research will utilize a survey questionnaire adapted from the study of Tolentino (2021) which includes the overall profile of the faculty researchers. This study will also integrate the components of the CAPSU RDE Training Office's Training Needs Assessment, which will expand the functionality of the stated survey questionnaire.

For the qualitative data, a semi-structured interview will be utilized to cater to the strengths, weaknesses, and difficulties of the faculty researchers within the institution. Lastly, primary data consisting of their research outputs, proposals, publications, citations, and other research-related outcomes will be analyzed in order for it to be contrasted side by side with the results of the first two data sets.

Data Collection

Permission from the Campus Administrators, Satellite College Directors, and the RDE Office will be requested prior to conducting the study. The selected participants will then be given a consent form along with the attached survey questionnaire. In turn, the results of the survey will be used to deduce the necessary questions for the qualitative interview.

After validating both quantitative and qualitative results, a letter of request will be sent to the RDE office to allow the researchers to procure the primary data from the faculty researchers for further analysis.

Respondents of the Study

This research will utilize Multi-Stage Sampling based on the standard set by the RDE Office for which they consider a faculty researcher of CAPSU which narrows it down to a set of key informants and participants. This will be the most appropriate sampling technique to attain the objectives of this study. In turn, this will improve the trustworthiness and timeliness of the study.

Table 1. Inclusion and Exclusion Criteria

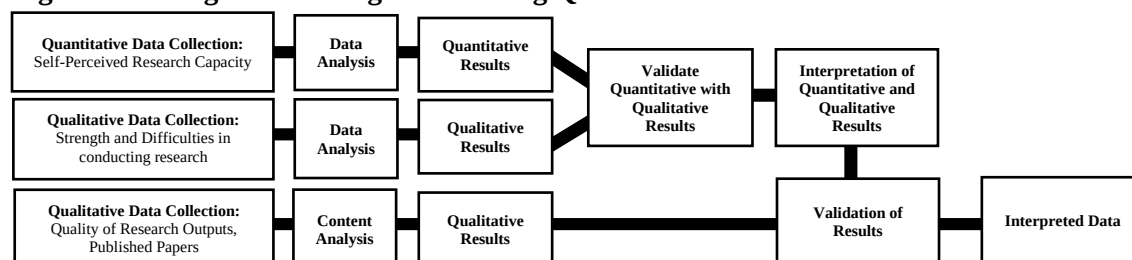
Inclusion Criteria	Exclusion Criteria
The participant must be a regular faculty that are required to do research (Associate Professor and Above)	Temporary personnel and emergency teachers as well as faculty researchers below Associate professor are not included in the study.
The researcher must have 1 or more completed research (University/Campus/Satellite College Level)	Researchers who were not able to produce 1 completed research will not be included in the study.
The researcher must have 2 or more ongoing research (University/Campus/Satellite College Level)	Researchers doesn't have 2 or more ongoing research will not be included in the study.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 6 of 8

Data Analysis Procedure

The data from the quantitative component will be processed in the SPSS software in order to extract the percentage and frequency from the survey questionnaire. The open-ended interview will then be processed through a thematic analysis and evaluated alongside its quantitative counterpart. The results of these components will then be assessed by the researchers with the results of the primary data evaluation.

Figure 1. Triangulation Design: Validating Quantitative Data Model Framework



(14) Major Activities/Work plan

	2022 Activity Workplan											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Requesting of Permissions and Gathering of Preliminary Data												
Field Work – Gathering of Quantitative Data												
Analysis of Quantitative Data												
Field Work – Gathering of Qualitative Data												
Analysis of Quantitative Data												
Gathering of Primary Data												
Content Analysis												
Triangulation of Data and Synthesis of Research Paper												
Presentation												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

1. Year-round capability-building program for the research capacity of the CAPSU faculty researchers.
2. Comprehensive graphic organizer of the CAPSU faculty researchers.
3. List of Software/Websites/Tools that will help in enhancing the research capacity of the faculty researchers.
4. Identified national/international partner agencies for collaboration in enhancing the research capacity of CAPSU as a whole.

(16) Targets Beneficiaries

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 7 of 8

Not Applicable
(17) Personnel Requirement
Not Applicable
(18) Budgetary Requirement
Training Expenses Supplies and Materials (kits, etc) - 2,000 Honorarium for Resource Persons Foods Expenses - 12,600 Representation - Communications - 1,000 Transportation & Gasoline - 18,000 Rental Expenses Printing and Binding - 3,000 Professional Fees - 1,500 Advertisement TOTAL - 38,100
(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/)
Abrazado, G & Coronel C. (2016). The research capability and workload profile of teachers vis-à-vis research productivity. The Philippine Journal of Graduate Education, Vol.2, No.1: February 2016 (pp.1-21), ISSN 2423-1401 Bowsher, G., Papamichail, A., El Achi, N., Ekzayez, A., Roberts, B., Sullivan, R., & Patel, P. (2019). A narrative review of health research capacity strengthening in low and middle-income countries: lessons for conflict-affected areas. <i>Globalization and health</i>, 15(1), 1-13. Capiz Quickstat - 2018. Philippine Statistics Authority. (n.d.). Retrieved November 17, 2022, from https://psa.gov.ph/content/capiz-quickstat-june-2018 Chu, K. M., Jayaraman, S., Kyamanywa, P., & Ntakiyiruta, G. (2014). Building research capacity in Africa: equity and global health collaborations. <i>PLoS medicine</i>, 11(3), e1001612. Creswell, J. W. (2014). <i>A concise introduction to mixed methods research</i>. SAGE publications. Dutta, S., Lanvin, B., León, L. R., & Wunsch-Vincent, S. (Eds.). (2021). Global innovation index 2021: tracking innovation through the covid-19 crisis. WIPO. Huenneke, L. F., Stearns, D. M., Martinez, J. D., & Laurila, K. (2017). Key strategies for building research capacity of university faculty members. <i>Innovative higher education</i>, 42(5), 421-435

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 8 of 8

Kotsemir, M., & Shashnov, S. (2017). Measuring, analysis and visualization of research capacity of university at the level of departments and staff members. *Scientometrics*, 112(3), 1659-1689.

Kurt, S. (2018). Why do authors publish in predatory journals?. *Learned Publishing*, 31(2), 141-147.

Overview of the position classification and compensation system. Department of Budget and Management. (n.d.). Retrieved November 16, 2022, from <https://www.dbm.gov.ph/wp-content/uploads/2012/03/Manual-on-PCC-Chapter-1.pdf>

Taherdoost, H. (2016). Sampling methods in research methodology; how to choose a sampling technique for research. How to choose a sampling technique for research (April 10, 2016).

Tolentino, K. S. (2021). The Research Capability Of Secondary School Science Teachers. *International Journal of Multidisciplinary: Applied Business and Education Research*, 2(3), 213-224.

VECALDO, R., Asuncion, J. E., & Mark, U. L. L. A. (2019). From writing to presenting and publishing research articles: Experiences of Philippine education faculty-researchers. *Eurasian Journal of Educational Research*, 19(81), 147-164.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)